MIL-P-19990A (WEP)
29 January 1964
Superseding
MIL-T-19990 (AER)
10 June 1957

MILITARY SPECIFICATION
PLOTTING TABLE, PHOTOGRAPHIC FILM

(This specification has been approved by the Bureau of Naval Weapons, Department of the Navy)

1. SCOPE

1.1 SCOPE. - This specification covers the requirements for an illuminator to be utilized for viewing and annotating photographic film negatives and positives. The Plotting Table shall be designated AR-6 ( ). (See 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on the date of invitation for bids shall form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-766	Steel Plates, Sheets, and Strips, Corrosion-resisting

Military

MIL-P-3757	Packaging of Photographic Accessories and Spare Parts Domestic & Overseas Shipment
MIL-C-3767	Connectors: Electrical, (Power Bladed Type), General Specification
MIL-D-70327	Drawings, Engineering and Associated Lists

FSC 6740

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MIL-M-9868	Microfilming of Engineering Documents 35mm, Requirements for
MIL-P-116	Preservation, Methods of

STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS-26565	Spool, Photographic Film, Aerial

(Copies of applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120. Attention: Code DCI)

2.2 OTHER PUBLICATIONS. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply).

NATIONAL BUREAU OF STANDARDS

Handbook H-28	Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402).

3. REQUIREMENTS

3.1 DESIGN AND CONSTRUCTION. - The plotting table shall be an internally illuminated rectangular enclosure with a flat, transparent, plastic top surface, and retractable legs. The plotting table shall provide for viewing 9 by 18 inch pictures on 9-1/2 inch wide aerial roll film, by transmitted illumination. The plotting table shall be constructed as a portable single unit to permit its use on a table top, with or without its legs

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extended. For operation, it shall require connection to an electrical source only. Wherever possible, commercially standard parts shall be used in construction of the plotting table. The plotting table shall be constructed to provide long periods of trouble-free operation.

3.2 No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order. (See 6.2).

3.3 MATERIALS. - Materials shall be as specified herein. Materials not definitely specified shall be of good commercial quality entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect the serviceability or appearance of the finished product.

3.4 CORROSION RESISTANCE. - The plotting table shall be made of inherently corrosion resistant materials or materials suitably treated to resist corrosion from damp salt air.

3.5 FUNGUS RESISTANCE. - Materials resistant to fungus growth shall be used to the maximum extent possible. All other materials shall be suitably treated to resist the growth of fungi.

3.6 WEIGHT. - The plotting table shall be of minimum weight consistent with the other requirements of this specification, but shall not exceed 35 pounds.

3.7 DIMENSIONS. - The dimensions of the lamp box proper (housing) shall be held to a minimum consistent with the other requirements and shall not exceed 30 by 14 inches. With the legs completely retracted, the top surface of the box shall be not more than 3 inches above the surface upon which the light table rests. The maximum dimensions stated shall not include the film wind brackets or support legs when extended.

3.8 INTERCHANGEABILITY. - All units and parts of the plotting table shall be interchangeable without modification, with corresponding units and parts of other plotting tables furnished on the same contract.

3.9 RIGIDITY AND ALIGNMENT. - Each part of the plotting table shall be of sufficient strength and rigidity to maintain smooth operation and proper alignment throughout the life of the unit.

3.10 SHOCK, INCLINATION AND VIBRATION. - The plotting table shall operate satisfactorily under conditions of shock, vibrati and inclination incident to handling and use on shipboard.

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3.11 SCREW THREADS. - Screw threads employed in fabrication of the plotting table shall conform to the National Bureau of Standards Handbook H-28 and shall be of commercial standard size. Coarse threads shall be used in soft metals. Sheet metal screws, unless provided with locking means, shall not be used.

3.12 ENVIRONMENTAL CONDITIONS

3.12.1 Operational. - The plotting table shall operate satisfactorily in any ambient temperature ranging from plus 50 to 100 F. and up to 90 percent relative humidity.

3.12.2 Storage. - The plotting table shall not be damaged by storage at any temperature ranging from minus 65 F to plus 160 F.

3.13 FILM DAMAGE. - The plotting table shall not scratch or otherwise damage the film base or emulsion in any manner during its operation.

3.14 PRINCIPAL PARTS. - The plotting table shall consist of the following principal parts:

(a) Housing
(b) Viewing Surface
(c) Light source
(d) Rewind brackets
(e) Support legs

3.15 HOUSING. - The housing shall be the main support of the plotting table. It shall enclose the light source to protect against dust, moisture and damage. It shall provide support for the legs, viewing surface and rewind brackets. The housing shall be constructed of aluminum alloy and shall not be connected to any part of the electrical circuit.

3.15.1 Hold Down Rollers. - Suitable, free-running, hold down rollers, shall be provided at each end of the plotting table to guide the film from the rewinds to approximate contact with the viewing surface. The rollers shall be approximately 3/8 inch in diameter and approximately 10 inches in length. The rollers shall be of Class 1 corrosion resisting steel, with a No. 8 mirror polish, in accordance with Specification QQ-S-766 or nickel plated brass. Each roller shall be hinged at one end and provided with a snap lock fastener at the opposite end, to facilitate insertion and removal of film without rewinding the complete roll.

3.15.2 Guard Plate. - The top surface of the housing shall be fitted with a suitable guard plate under each hold down roller, to minimize the possibility of damage to the film as it passes from the film spools to the viewing surface. The guard plate

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shall be of Class 1, corrosion resisting steel, with a No. 8 mirror polish, in accordance with Specification QQ-S-766 or nickel plated brass.

3.15.3 Top. - If the top of the housing is hinged to allow access to the interior, the hinged portion shall be provided with a safety stop to prevent damage to the hinges from opening to an excessive angle.

3.16 VIEWING SURFACE. - The viewing surface shall be a plastic covered opening in the top of the housing for viewing and annotating film transparencies. The plastic covering shall consist of a clear top sheet as a working surface, and a translucent sub-sheet if required for diffusion of the light. The plastic sheets shall be recessed and properly secured in the housing but shall be easily removable. The top sheet shall be flush with the top surface of the housing. The top plastic sheet shall be so made that it can be reversed, bottom for top, when the working surface becomes scratched. The viewing surface shall be capable of supporting a viewing instrument weighing 20 pounds and mounted on a base approximately 7 inches in diameter, when tested as specified in Section 4. The viewing surface shall not sag, warp, or crack from expansion or heat generated within the light table, under the operational or environmental conditions specified herein. Means employed to prevent damage as described above, shall not cast shadows or otherwise affect the evenness of illumination at the viewing surface.

3.16.1 Viewing Area. - The viewing area shall be a full 9 by 18 inch illuminated opening, without obstructions of any kind.

3.16.2 Temperature Rise. - The temperature rise at the viewing surface shall not damage nitrate or acetate base film when viewed continuously over a 4 hour period. When tested as specified in Section 4, the temperature of the viewing surface shall rise not more than 30 F above ambient temperature.

3.16.3 Illumination. - The light at the viewing surface shall be adjustable for full intensity and low intensity, which shall be approximately one half of full intensity. When tested as specified in Section 4, full intensity shall provide not less than 530 foot/candles at any point in the viewing area. When determined as specified in Section 4, the intensity of any point in the viewing area shall not vary by more than 25 percent from the average intensity.

3.17 LIGHT SOURCE. - The light source or sources shall be contained within the housing. Each lamp assembly shall be of a rugged type, mounted in a manner to withstand rough handling of

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the plotting table. The color of light produced shall make an approximate visual match to a source of 4200 degrees Kelvin.

3.18 FILM REWIND BRACKETS. - Adjustable rewind brackets shall be provided at each end of the plotting table. The brackets shall be rigidly attached to the table but shall be readily removable. The brackets shall have manually operable rewind cranks to facilitate forward or backward winding of film across the plotting table. The rewind brackets shall accommodate film spools conforming to MS-26565 in widths up to 9-1/2 inches and lengths up to 500 feet. The design and mounting of the brackets shall permit rewinding the 500 foot spool with the plotting table on a flat surface and its legs not extended. Means shall be provided for lowering the film spools so that no portion of the flange extends above the plane of the viewing surface or contacts the supporting table surface when the legs are fully extended. Winding and rewinding of the film shall be possible when the film spools are in this position. Spring loaded spindles shall be provided to facilitate easy insertion and removal of film spools.

3.19 SUPPORT LEGS. - The plotting table shall have four support legs of corrosion resisting metal. The legs shall be of the captive type, capable of being stowed within or recessed into the table housing. The legs, when fully extended shall raise the viewing surface to a distance of 6-3/4 inches, plus or minus 1/8 inch from the surface of the supporting table. The front legs shall be adjustable throughout their length. Means shall be provided for locking the legs rigidly in position.

3.20 HAND GRIPS. - Recessed hand grips shall be provided in the ends of the plotting table for lifting and transport purposes.

3.21 ELECTRICAL REQUIREMENTS. - The plotting table shall operate satisfactorily from a 115 volt, 60 cycle, single phase AC electrical power supply.

3.21.1 Switches. - Recessed ON-OFF power and HIGH-LOW switches shall be located in the front of the plotting table.

3.21.2 Wiring. - All electrical wiring of the plotting table shall conform to standards of the Underwriters Laboratories.

3.21.3 Power Cord. - The plotting table shall be provided with a 10 foot length of rubber covered, 3 wire, grounding type power cord. One end shall be permanently attached to the light table and the other end shall terminate in a 3 prong male rubber plug which is in accordance with MIL-C-3767/4.

3.22 FINISH

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3.22.1 Interior. - The interior of the lamp housing, including fittings and wiring, shall be given a durable, smooth, white matte finish.

3.22.2 Exterior. - The exterior of the lamp housing shall be given a durable smooth grey finish.

3.23 SPARE PARTS. - Spare parts to be furnished with the plotting table shall be as specified in the procurement document (see 6.2).

3.24 IDENTIFICATION MARKING. - Identification marking of the equipment shall be in accordance with MIL-STD-130.

3.25 DRAWINGS. - When specified in the procurement document (see 6.2) a complete set of Class 2 drawings and all associated lists conforming to Specification MIL-D-70327 shall be furnished.

3.26 MICROFILM. - When specified in the procurement document (see 6.2) microfilm conforming to MIL-M-9868 shall be made of the drawings and associated lists.

3.27 WORKMANSHIP. - Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION. - Unless otherwise specified in the contract or purchase order (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, (see 6.2), the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 TEST CONDITIONS. - Unless otherwise specified herein, tests shall be conducted under ordinary room conditions of temperature and humidity.

4.3 TEST EQUIPMENT. - The equipment necessary for the simulation of actual use conditions for testing shall be furnished by the contractor.

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TABLE I

EXAMINATION AND TESTS

TESTS	REQUIREMENTS	TEST METHOD
Preproduction Inspection		
Visual and mechanical	3.1	4.5.1
Corrosion resistance	3.4	4.5.1
Fungus resistance	3.5	4.5.1
Interchangeability	3.8	4.5.1
Rigidity and alignment	3.9	4.5.5
Shock, vibration	3.10	4.5.5
Operating (environmental)	3.12.1	4.5.7
Film damage	3.13	4.5.7
Temperature rise	3.16.2	4.5.4
Illumination	3.16.3	4.5.2
Illuminants	3.17	4.5.3
Rewind brackets	3.18	4.5.1
Switches	3.21.1	4.5.1
Power cord	3.21.3	4.5.1
Quality Conformance Inspection		
Physical examination	3.1	4.6.1
Operation		4.6.2
Packaging & Packing Inspections		
Packaging	5.1.1 & 5.1.2	
Packing	5.1.2 & 5.2.2	
Marking	5.3	

4.4 CLASSIFICATION OF INSPECTIONS. - Inspections shall be classified as follows:

(a) Preproduction Inspection
(b) Quality Conformance Inspection

4.5 PREPRODUCTION INSPECTION. - The preproduction sample, prior to submission for approval, shall be subjected by the contractor to the preproduction tests specified herein, at the place of manufacture or at a test facility acceptable to the Government. At the option of the Bureau of Naval Weapons, any or all of the preproduction tests shall be monitored by a representative of the Bureau of Naval Weapons. On completion of the tests, the preproduction sample unit and a report of the results of the tests, when specified in the contract or order, shall be shipped to the Military Testing Agency designated by the Bureau of Naval Weapons (see 6.2). The preproduction tests shall consist of the inspections listed in Table I. Approval by the Bureau of Naval Weapons shall be based on no defects in the preproduction sample.

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4.5.1 Physical Examination. - Examine the plotting table visually and with gauges to determine conformance with the requirements of this specification with respect to design, materials, construction, wiring, dimensions, weight, components, finish and workmanship.

4.5.2 Illumination Tests. - Using a laboratory quality illumination meter, or equal, measure the light emitted from various parts of the viewing area with the controls set for high intensity. Measure the intensity one inch in from the sides of the viewing area at the four corners, at the mid-points of the four sides, the center and at the four mid-points between center and corners along the diagonals. Visually examine the viewing area for local bright spots or shadows. If any are observed, measure the brightness of that area. No reading taken shall differ by more than 25 percent from this average. Repeat the test with the controls set for low intensity. The average reading shall be between 200 and 320 and no reading shall differ by more than 25 percent from this average.

4.5.3 Color of Light. - Compare the color of the viewing surface of the plotting table at high intensity with a diffuse light source having a color temperature of 4200 Kelvin. The appearance of the two shall be approximately the same.

4.5.4 Temperature Rise. - Attach a wire thermocouple to the center of the viewing surface with transparent plastic tape and on it lay either a 9 by 18 inch picture negative bearing the usual distribution of densities or a negative uniformly fogged and developed to a density of 0.9 plus or minus 0.1. Operate the plotting table at an ambient temperature of 75 plus or minus 10 F. until the temperature stabilizes. The temperature rise at the surface shall not exceed 30 F. above ambient.

4.5.5 Viewing Surface Rigidity. - Establish a plane parallel to the viewing surface of the plotting table and with suitable gauges measure the distance from the datum plane to the plastic surface at existing room temperature. Place a circular object seven inches in diameter with a central opening of suitable size and weighing twenty pounds, in the center of the viewing surface. Operate the plotting table at high intensity in an ambient temperature of 105 F for 20 hours. Without changing conditions, measure the distance from the datum plane to the center of the viewing surface. The distance shall not have changed by more than 0.1 inch.

4.5.6 Viewing Surface Warping. - Establish a plane parallel to the viewing surface of the plotting table and with suitable gauges measure the distance from the datum plane to the plastic surface at room temperature. Raise the ambient temperature to

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160 F and maintain for four hours. Allow the plotting table to cool to room temperature. Remeasure the distance from the datum plane to the plastic surface. The distance to any point on the viewing surface shall not have changed by more than 0.05 inch. Visually examine the plastic surface sheet and diffuser. No indication of damage shall be perceptible.

4.5.7 Operation. - Adjust the legs at various settings and determine that the viewing surface is held firmly at the various angles. Determine that the various size spools fit easily and securely in the rewinds. Load a spool with 500 feet of 9-1/2 inch wide film on the plotting table and determine that the rewinds operate satisfactorily in the raised and lowered position while the viewing surface is set at various angles. Examine the film after several rewindings to determine that it has not been scratched or damaged.

4.6 QUALITY CONFORMANCE INSPECTION. - Each plotting table, after the Preproduction Sample, offered for acceptance on the contract or order shall undergo Quality Conformance Tests conducted by the contractor at the place of manufacture, under supervision of the Government Inspector.

4.6.1 Physical Examination. - Examine the plotting table visually to determine that it matches the preproduction sample and conforms with the requirements of this specification with respect to fit of components, wiring, finish and workmanship.

4.6.2 Operation. - Operate the controls, rewinds and legs to determine that all lamps light and that the moving parts and locks function properly.

4.7 PACKAGING AND PACKING. - Examine the method of preservation, packaging, packing and marking for shipment and storage visually and mechanically to determine conformance to the requirements of the applicable specifications.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING. - Preservation and packaging shall be Level A as defined in Federal Standard No. 102. The unit shall be packaged as one integral unit.

5.1.1 Level A. - Units shall be preserved and packaged in accordance with the procedures specified in Specification MIL-P-116 using Method 1A8. No contact preservation shall be required.

5.1.2 Level C. - Units shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

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5.2 PACKING. - The levels of packing shall be as specified by the procuring activity (see 6.2).

5.2.1 Level A. - Unit packages shall be packed for shipment in accordance with Specification MIL-P-3757. No more than twenty unit packages shall be packed in any one shipping container.

5.2.2 Level C. - Unit packages shall be packed for shipment in accordance with Specification MIL-P-3757. No more than twenty unit packages shall be packed in any one wood shipping container. If fiberboard shipping containers are used, the number of unit packages shall be such as to insure safe delivery to the point of acceptance.

5.3 MARKING. - All unit packages and shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 INTENDED USE. - The plotting table is intended for viewing negative and positive film transparencies by trans-illumination and to provide a flat, smooth surface for making annotations, overlays and tracings.

6.2 ORDERING DATA. - Procurement documents should specify the following:

(a) Title, number and date of the specification
(b) Quantity of units desired.
(c) Responsibility for inspection (see 4.1)
(d) Preproduction inspection (see 4.5)
(e) Level of preservation, packaging and packing (see Section 5)
(f) Drawings, if required (see 3.25)
(g) Microfilm, if required (see 3.26)
(h) Spare parts (see 3.23)
(i) Reports, if required (see 4.5)

6.3 OMISSION OF PREPRODUCTION TESTS. - If the unit has successfully passed the preproduction tests during the past twelve months, preproduction testing of additional identical units from the same manufacturer may be waived. Authorization to omit the preproduction tests may be requested, by the contractor, from the requiring agency.

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6.4 TYPE DESIGNATOR. - In accordance with the provisions of MIL-STD-155, the parenthesis in the type designator will be replaced by a letter, number, or both. This information will be furnished upon application by the contractor to the requiring activity after award of each contract. The complete type designator shall then be used on nameplates, in instruction books, and wherever reference is made to the equipment covered by this specification.

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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NAVY DEPARTMENT

Chief, Bureau of Naval Weapons (RREN-5)

Navy Department

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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